جامعة بجاية
Tasdawit n'Bgayet
Université de Béjaïa

Computer Science Department
First year of Bachelor's degree

Chapter 2 - AI for Document Retrieval (3)

https://zair-bouzidi.github.io/zair-site/

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Introduction

- **Literature review** is a **key step** in any **scientific project**.

With the rise of **AI**, it is **now** possible to automate and optimize **many tasks** related to the **collection**, **analysis** & **synthesis** of scientific information.

This chapter :

- **presents** **AI-based tools**,
- **explains** **how to write intelligent** queries,
- automatically **summarize** **articles**, and
- critically **evaluate** the **sources suggested** by these **tools**.

Presentation of AI tools

Several platforms use **AI** to facilitate **documentary research** :

Elicit : allows you to automatically **extract** key information from scientific publications, **generate** summaries, and **suggest** relevant articles

Scopus AI : Integrated into the Scopus database, it helps **analyze** publications, **identify** scientific trends, and **recommend** articles for further exploration of a topic

Web of Science Research Assistant : This intelligent assistant **suggests** relevant articles, **creates** reference lists, and **allows** you to quickly filter results based on **quality**, **impact**, and **relevance**

Designing Smart Requests

To get the most out of **AI tools**, it is essential to know how to write **effective queries** :

- **Define** **precise** & **relevant** **keywords** for the **research topic**

- **Use** **advanced filters** (dates, publication types, scientific field)

- **Combine** **keywords** with **logical operators** (**AND**, **OR**, **NOT**) to **refine** the results

- **Test** different **formulations** & **adjust** the **queries** based on the **results obtained**

Automated Abstract of Scientific Articles

AI makes it possible to **quickly generate summaries** of **large articles** :

- **Identification** of **key points** and **main conclusions**

- **Creating** **comparative summaries** when several articles deal with the same subject

- **Saving** **time** for **researchers** wishing to focus on **critical analysis** rather than exhaustive reading

Critical Evaluation of AI Proposed Sources

Even though **AI** is very useful, it's **important** to **verify** the **quality** of the **sources**:

- **Check** the **reliability** and **credibility** of the **suggested articles**

- **Analyze** the **potential biases** of the **AI tool** (e.g., some platforms favor certain publishers or disciplines)

- **Select** only the **relevant** & **rigorous sources** for the **scientific project**

Conclusion

AI is **transforming** **information retrieval** by offering **powerful tools** for information gathering and analysis.

However, the researcher's role **remains** **crucial** in

- **formulating** the **right queries**,
- **interpreting** **results**, and
- **selecting** **reliable sources**.

By combining **AI** with a **critical approach**, it is possible to significantly **improve** the **quality** & **efficiency** of **scientific research**.

How use Elicit?

What is Elicit?

Elicit is an **AI Tool** that **enables** to :

- **Retrieve Relevent Scientific Articles**

- **Abstract Publications**

- **Extract Key information** :
    * **Methodology**,
    * **Results**,
    * **Sample**,
    * **etc**.

- **Automatically compare** many **studies**

Unlike Google Scholar, **Elicit** does more than just display links :

☞ it **analyses Contents** of **Articles** and **Extracts Important Information**

Steps of Utilisation

Step 1 : Formulate a Net Question

Elicit method **works** **best** with a **specific question**, and not just with **keywords**

Bad Example:

- Artificial Intelligence education

Good Example:

- What is the impact of artificial intelligence on student academic performance in higher education?

Step 2 : Launching the Research

While input the question :

- **Elicit** **displays** a List of **Relevent Scientific Articles**, **generating** a table containing :
  - Title
  - Year
  - Type of study
  - Methodology
  - Main Results
  - Automated Abstract

Step 3 : Analysizing & Comparing

We can:

- Add Columns (such as: sample size)

- Sort studies by date

- Comparing the results

- Export data

Concrete Example

Imagine you are preparing an Article :

The impact of deep learning on epidemiological surveillance

In **Elicit**, we put the following question :

How does deep learning improve infectious disease surveillance systems?

Article	Method	Data	Main Result
Study A (2023)	CNN	Twitter data	Early detection 2 weeks before official reports
Study B (2022)	LSTM	Hospital data	18% improvement in accuracy
Study C (2021)	Hybrid DL	Multi-source data	Reduction of the alert period

In some minutes, we obtain:

- A Comparative view
- Methodological trends
- Calculated performance
- The limitations mentioned

Without reading all 30 articles manually

Why it is Stong for Researchers ?

- ✓ Huge time saving
- ✓ Can Structure a Literature Review
- ✓ Enabling quick identification of gaps
- ✓ Ideal for preparing an introduction or a state-of-the-art review

Important Limitations

- ✓ **Elicit** does not replace reading the articles in their entirety
- ✓ It can summarize in simplified terms
- ✓ **It must always check the original sources**

PS:

Always Check the Original Sources

Practical Exercise

Using Elicit for AI-assisted scientific research

Educational objectives

At the end of this exercise, the student will be able to :

- **Formulate** a precise research question
- **Use** **Elicit** to identify relevant scientific articles
- **Compare** multiple studies
- **Summarize** and critically **analyze** results
- **Evaluate** the reliability of sources suggested by AI

Part 1: Formulation of the question

Instructions

Choose a theme related to artificial intelligence

Examples of themes

- AI and academic performance

Deep learning and public health

AI and climate change

Generative AI and education

Formulate a clear scientific question

PS: Formulate a clear scientific question

Formulate a clear scientific question

❌ **Bad example:**

- Artificial Intelligence in health

✅ **Good example**

- How does deep learning improve early detection of infectious diseases?

✍ **Task**

- Write the question
- Justify why it is clear and precise (5–6 lines)

Part 2: Search with Elicit

Instructions

1. Log in to Elicit (www.Elicit.com)

2. Enter your question

3. Select the 5 most relevant articles

Complete the following table

Article	Year	Used Method	Data Type	Main Result	Limitations

Part 3: Comparative Analysis

Answer the following questions :

1. **Which AI method is most frequently used?**
2. **Are the results consistent across studies?**
3. **Are the sample sizes sufficient?**
4. **Are there any contradictions?**
5. **Which study seems the most robust? Why?**

(1-2 pages maximum)

Part 4: Critical evaluation of AI

Answer :

1. Are the summaries generated by Elicit complete?

2. Did you find any missing information after checking?

3. Did Elicit suggest any irrelevant articles?

4. Can the tool be trusted blindly? Justify your answer.

Part 5: Mini-Summary

Write a 300–400 word summary:

- **Summarize the main trends**
- **Indicate the identified limitations**
- **Suggest a direction for future research**

Evaluation methods (example)

Criteria	Points
Quality of the question	/4
Relevance of the selected articles	/4
Comparative analysis	/6
Critical thionine	/4
Editorial quality	/2
Total	/20

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Prolog
Basic Concepts

COURSE of PROLOG: Formalization

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% --- Facts : who like what ---

Adam likes Apples.

like(adam, apples).

Ines likes carrots.

like(ines, carrots).

Karim likes oranges.

like(karim, oranges).

% --- Facts : categories ---

Apples are fruits.
fruit(apples).

Oranges are fruits.
fruit(oranges).

Carrots are vegetables.
vegetable(carrots).

```
% --- Facts : categories ---
fruit(apples).
fruit(oranges).
vegetable(carrots).

% --- Rule : being healthy ---
healthy(Person) :- like(Person, Food),
                   fruit(Food).
```

A person is healthy if he likes a food AND that food is a fruit.

🔎 Explanation of the rule

```
healthy(Person):-
like(Person, Food),
fruit(Food).
```

This means:
A person is healthy if he likes a food AND that food is a fruit.

✅ 2 Query: Who is healthy?
?- healthy(X).

✔ Expected result:
X = Adam;
X = Karim.

Why?
Adam likes apples → fruit ✅
Karim likes oranges → fruit ✅
Inès likes carrots → vegetable ❌

✅ 3 Query: Who likes apples?

This allows us to ask:
?- food(X).

🎯 Pedagogical Summary

Prolog Query Question
Who is healthy? - healthy(X).
Who likes apples? - like(X, apples).
What are some fruits? - fruit(X).

?- likes(X, apples).

✔ Result:

X = Adam.

✅ 4 How to find out which fruits are known from the curriculum?

How to find out which fruits are known from the curriculum?

?- fruit(X).

✔ Result:

X = apples;

X = oranges.

🎓 Slightly more general version (optional)

We can also define:

food(X) :- fruit(X).

food(X) :- vegetable(X).

Statement:

- Adam likes apples.
- Inès likes carrots.
- Karim likes oranges.
- Apples are fruits.
- Oranges are fruits.
- Carrots are vegetables.
- Those who like fruit are healthy.

Questions:

1. Formalize these facts and rules in PROLOG.
2. What is the query to find out who is healthy?
3. What is the query to find out who likes apples?
4. How can we find out which fruits the program knows?